
Amazon Athena – Lab Part I: Query CSV from S3

Objective:

Query a sample CSV file from S3 using Athena + Glue Catalog.

Step 1: Upload CSV File to S3

1. Go to S3 → Create a bucket (e.g., `athena-lab-demo`)
2. Upload `sales.csv`:

```
order_id,product,category,amount,region
1001,Laptop,Electronics,75000,South
1002,TV,Electronics,50000,North
1003,Shoes,Fashion,3000,East
```

Step 2: Create AWS Glue Crawler

1. Go to AWS Glue → Crawlers → Create Crawler
 2. Data Store: S3 → Path: `s3://athena-lab-demo/`
 3. Output Database: Create new (e.g., `athena_lab_db`)
 4. Run the crawler → A table (e.g., `sales_data`) appears
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Step 3: Query from Athena

1. Go to Athena console

2. Select the database `athena_lab_db`

3. Run:

```
SELECT * FROM sales_data;
```

4. Preview the results and explore queries like:

```
SELECT region, SUM(amount) AS total_sales  
FROM sales_data  
GROUP BY region;
```

Amazon Athena – Lab Part II: Query Partitioned Parquet Data

Objective:

Query Parquet data partitioned by region for better performance and cost.

Step 1: Convert and Upload Partitioned Data

Upload Parquet files structured as:

```
s3://athena-lab-demo/region=South/sales.parquet  
s3://athena-lab-demo/region=North/sales.parquet
```

Step 2: Create Partitioned Table in Athena

```
CREATE EXTERNAL TABLE IF NOT EXISTS sales_partitioned (  
  order_id INT,  
  product STRING,  
  category STRING,  
  amount INT  
)  
PARTITIONED BY (region STRING)
```

STORED AS PARQUET
LOCATION 's3://athena-lab-demo/';

✓ **Step 3: Load Partitions**

MSCK REPAIR TABLE sales_partitioned;

✓ **Step 4: Run Optimized Query**

```
SELECT region, COUNT(*) AS order_count  
FROM sales_partitioned  
WHERE region = 'North'  
GROUP BY region;
```

✓ Athena scans only relevant partitions → saves cost!
